

ORIGINAL CONTRIBUTION

Colchicine for Secondary Prevention After Stroke According to the Presence of Atherosclerosis

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BACKGROUND: Anti-inflammatory therapies reduce the risk of major adverse cardiovascular events (MACEs) in coronary disease, but efficacy has not been shown after stroke. It is uncertain if patients with atherosclerosis are more likely to benefit from anti-inflammatory prevention.

METHODS: We performed a post hoc analysis of the CONVINCe trial (randomized, open-label, blinded-end point assessed; Colchicine for Prevention of Vascular Inflammation in Non-Cardioembolic Stroke), comparing colchicine 0.5 mg plus usual care versus usual care. We performed a subgroup analysis in patients stratified according to the presence/absence of atherosclerosis (defined as cervico-cranial/aortic artery plaque, coronary disease, peripheral arterial disease, and carotid revascularization). The primary outcome was MACE. Analyses were adjusted for age, qualifying event, time from event to randomization, and coronary disease.

RESULTS: Three thousand one hundred forty-four participants were included. There were 338 MACE events during follow-up (median, 33.6 months). On intention-to-treat analysis, there were numerically fewer MACE events in the atherosclerosis group assigned to colchicine (105 [11.1%], 3.88 [95% CI, 3.21–4.70] per 100 person-years) versus usual care (132 [14.2%], 4.83 [4.07–5.73] per 100 person-years), which was not statistically significant (hazard ratio [HR], 0.83 [95% CI, 0.64–1.07]). There was no difference in the rate of the primary outcome according to treatment allocation in the nonatherosclerosis subgroup (HR, 0.94 [95% CI, 0.64–1.39]; $P_{\text{interaction}}=0.60$). There was a nonsignificant reduced risk of recurrent ischemic stroke in the colchicine arm (HR, 0.76 [95% CI, 0.56–1.05]) in the atherosclerosis group, but no difference in events was observed in the nonatherosclerotic group (HR, 0.92 [95% CI, 0.60–1.42]; $P_{\text{interaction}}=0.48$). In the prespecified on-treatment analysis, colchicine reduced the risk of MACE in patients with atherosclerosis (HR, 0.76 [95% CI, 0.58–0.99]) but not without (HR, 0.95 [95% CI, 0.63–1.42]; $P_{\text{interaction}}=0.37$).

CONCLUSIONS: There were no significant between-group interactions for colchicine and atherosclerosis for MACE in intention-to-treat or on-treatment populations. However, outcome events were numerically fewer in patients with atherosclerosis assigned to colchicine, and significant benefit was observed in the on-treatment analysis in this group. Data suggest that the presence of atherosclerosis should be an inclusion criterion in future anti-inflammatory therapy trials.

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Key Words: atherosclerosis ■ inflammation ■ prognosis ■ randomized controlled trial ■ stroke

Globally, there are over 90 million people living after stroke, and each year, there are 12 million new strokes.¹ Despite effective secondary prevention

therapies, the 5-year rates of recurrent stroke or major coronary events are 25% to 30%.² Most disability and death attributed to stroke occurs in low-income and

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Nonstandard Abbreviations and Acronyms

aHR	adjusted hazard ratio
CAD	coronary artery disease
CONVINCE	Colchicine for Prevention of Vascular Inflammation in Non-Cardioembolic Stroke
CRP	C-reactive protein
HR	hazard ratio
IL	interleukin
ITT	intention-to-treat
MACE	major adverse cardiovascular event
NLRP3	NLR family pyrin domain-containing 3
RCT	randomized controlled trial

low-middle-income countries.¹ To address this global challenge, there is an urgent need for new stroke prevention therapies that will be effective from both clinical and economic perspectives.

Inflammation plays a pivotal role at all stages of atherosclerotic plaque development.³ There is also compelling evidence arising from clinicopathological studies,^{4,5} epidemiological data,^{6–9} and Mendelian randomization^{10–12} studies that inflammatory mechanisms are central to the pathogenesis of ischemic stroke. However, ischemic stroke is heterogeneous from an etiological standpoint, with multiple and often overlapping disease pathologies implicated, including atherosclerosis, cardioembolism, and cerebral small vessel disease.^{13,14} While there is convincing evidence for inflammation in atherosclerosis, the relative contribution to cardioembolism and cerebral small vessel disease is less certain. Moreover, patients with atherosclerotic stroke seem to be at higher risk of long-term vascular recurrence compared with patients without atherosclerosis and may, therefore, derive greater absolute benefit from effective anti-inflammatory therapies.¹⁵ As inflammation evolves as a candidate target for stroke prevention, it is unclear whether benefit from anti-inflammatory therapies will be primarily in patients with atherosclerosis.

Colchicine is an inexpensive, widely available drug used in the treatment of acute gout and recently demonstrated efficacy as a long-term anti-inflammatory therapy for reducing major adverse cardiovascular events (MACEs) in coronary artery disease (CAD) in several randomized controlled trials (RCTs).¹⁶ However, a recent RCT of long-term colchicine and another RCT of acute short-term colchicine did not show evidence of benefit for secondary prevention after ischemic stroke.^{17,18} It has been suggested that one of the reasons for the neutral findings of these RCTs was the broad selection strategy, which included patients with and without atherosclerosis. Future RCTs of anti-inflammatory therapies after stroke are planned, and evidence to address this question is important in the design of such trials.

Here, we present a post hoc analysis of the CONVINCE trial (Colchicine for Prevention of Vascular Inflammation in Non-Cardioembolic Stroke), which evaluated the relative risk of vascular recurrence associated with atherosclerosis and analyzed the treatment effect of colchicine in subgroups defined according to the presence or absence of atherosclerosis.

METHODS

Data Sharing

Data are available upon reasonable request.

Trial Design and Eligibility

CONVINCE was a parallel-group prospective, randomized, open-label, blinded-end point–assessed phase 3 clinical trial, details of which have been previously published.^{17,19} The trial was conducted in 144 hospital sites from 14 countries from December 2016 to November 2022. After written informed consent, participants were randomized 1:1 to either colchicine plus guideline-recommended usual care or usual care alone using a Web-based computerized algorithm, which included minimization variables to ensure that groups were balanced for key prognostic variables: age (<70 or ≥70 years), time since qualifying event (<7 or ≥7 days), and qualifying event (stroke or transient ischemic attack).

The inclusion criteria were (1) age ≥40 years; (2) noncardioembolic stroke or high-risk transient ischemic attack; and (3) randomization between 72 hours and 28 days from qualifying event. Key exclusion criteria included cardioembolic stroke or other defined mechanisms (eg, vasculitis), moderate-to-severe renal, liver, or blood disorders, and estimated glomerular filtration rate <50 mL/min.

Colchicine was administered at a low dose of 0.5 mg orally per day. Participants, site investigators, and treating clinicians were aware of treatment allocation. Results were reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology reporting guideline.²⁰ The trial was approved by institutional review boards in each participating hospital (the first ethics approval was by the Mater University Hospital Dublin, reference 1/478/75).

Follow-Up and Outcome Adjudication Procedures

Follow-up visits occurred at 28 days, 90 days, and 6-month intervals from randomization, either by telephone or in-person. Suspected outcome events were assessed by an independent adjudication committee masked to the treatment allocation, comprising 2 stroke physicians and 2 cardiologists. Agreement between adjudicators was required for outcome verification. In the event of discordant adjudication, the case was assessed by a third independent blinded adjudicator and outcome status decided by the majority opinion.

Outcome Definitions

The primary outcome was first MACE during follow-up, defined as a composite of nonfatal ischemic stroke, nonfatal major coronary event, or vascular death. The prespecified key secondary

end point was the composite of nonfatal recurrent ischemic stroke, myocardial infarction, cardiac arrest, or vascular death. Other secondary end points included any recurrent ischemic stroke, any major coronary event, and vascular death.

Atherosclerosis Status

For this analysis, atherosclerosis was defined as evidence of ≥ 1 of the following before randomization: (1) qualifying stroke/transient ischemic attack due to symptomatic cranio-cervical stenosing ($>50\%$) atherosclerotic plaque; (2) stroke ipsilateral to aortic arch or nonstenosing cranio-cervical plaque ($<50\%$) irrespective of the presence of an alternative stroke etiology; (3) asymptomatic cranio-cervical plaque with stenosis $>50\%$; (4) history of carotid endarterectomy or stenting; (5) history of CAD (previous myocardial infarction, percutaneous coronary intervention, or coronary artery bypass graft); or (6) history of peripheral artery disease. Patients with missing data for ≥ 1 of these criteria were not excluded from the main analysis, and atherosclerosis status was assigned based on available data.

Statistical Analysis

Continuous variables are presented as mean (SDs) or median (interquartile intervals). Categorical variables are presented as number (%). Differences in baseline characteristics between patients with and without atherosclerosis were analyzed using the χ^2 test, the Mann-Whitney U test, or t tests as appropriate. We assessed whether baseline characteristics differed according to treatment allocation when patients were stratified according to atherosclerosis status and calculated the rate of recurrent events in atherosclerosis subgroups. The primary analysis was performed by intention-to-treat (ITT), with censoring at the time of first MACE, death, or last follow-up (whichever came first). We performed a prespecified on-treatment analysis, excluding patients who never commenced colchicine and with additional censoring for last-known adherence to colchicine in participants who permanently discontinued the trial drug. Cox proportional hazards regression modeled the effect of the intervention compared with usual care alone, stratified according to atherosclerosis status. We adjusted for mandatory randomization minimization variables (age <70 versus ≥ 70 years, time from event to randomization, and index event) and any baseline characteristics, which differed according to treatment allocation ($P<0.05$). The effect modification of colchicine by atherosclerosis subgroup was formally assessed using a test for interaction. A sensitivity analysis was conducted, which excluded patients with missing data for vascular imaging, who otherwise did not meet the criteria for atherosclerosis. All analyses were performed using Stata Release 17.

RESULTS

Study Population

After the exclusion of 10 patients who withdrew consent for use of their data, the trial included 3144 patients; 1875 (59.6%) participants had evidence of atherosclerosis (Figure 1). The baseline characteristics according

to atherosclerosis status are presented in Table 1. When further stratified according to treatment assignment (Tables S1 and S2), there were no significant differences in baseline characteristics apart from a slightly higher proportion of patients with CAD in the atherosclerosis group allocated to usual care compared with those assigned to the colchicine arm. There were high rates of antithrombotic and statin use in both groups irrespective of atherosclerosis status. There was no missing data prior to CAD, peripheral artery disease, or carotid revascularization. Vascular imaging was not performed or was uninterpretable in 243 patients, of whom 31 were still assigned to the atherosclerosis subgroup as they had met at least 1 of the other criteria.

Follow-Up

During a cumulative follow-up period of 9326.5 years, there were 338 first MACE outcomes during follow-up, of which 234 were due to nonfatal stroke, 55 due to nonfatal myocardial infarction, 14 due to hospitalization for unstable angina, 2 due to nonfatal cardiac arrest, and 33 due to vascular death (10 fatal strokes and 23 fatal coronary events). A total of 321 (20.5%) of participants assigned to colchicine were nonadherent during follow-up. Where available, reasons for discontinuation were adverse events in 138 (44.0%), patient decision to discontinue in 100 (31.8%), non-adherence in 23 (7.3%), and other reasons in 53 (16.9%).

Recurrence Rates According to Atherosclerosis Status

The risk of vascular recurrence was substantially higher in patients with evidence of atherosclerosis. The crude rate of MACE was 4.36 (95% CI, 3.84–4.95) events/100 person-years in patients with atherosclerosis and 2.60 (95% CI, 2.14–3.16) events/100 person-years in those without atherosclerosis (Figure 2). The corresponding rates for recurrent stroke were 2.92 (95% CI, 2.50–3.42) and 2.19 (95% CI, 1.77–2.70) events/100 person-years in participants with and without atherosclerosis, respectively (Figure S1). The rates of major coronary events were similarly higher in the atherosclerosis group, approaching 1.5%/y compared with a baseline rate of just 0.4%/y in the nonatherosclerosis group (Figure S2). Hazard ratios (HRs) for outcome events according to the presence of atherosclerosis versus no atherosclerosis were MACE, 1.66 (95% CI, 1.32–2.10), recurrent stroke, 1.32 (95% CI, 1.01–1.71), and major cardiac events, 3.52 (95% CI, 2.06–6.03). A similar increased risk of recurrence was observed when the atherosclerosis group was further stratified using TOAST criteria for large artery atherosclerotic stroke (Table S3; Figure S3).

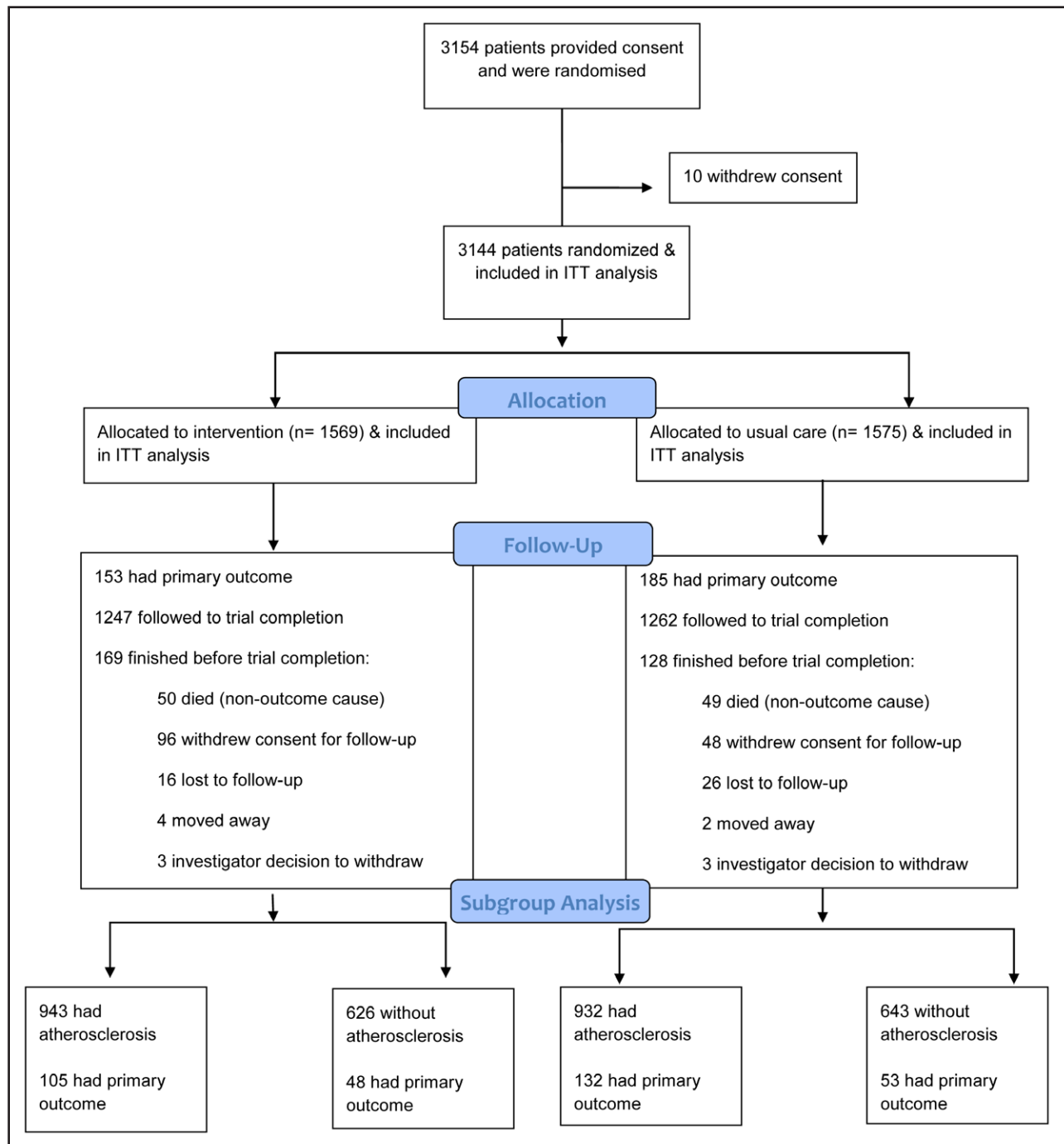


Figure 1. Flow diagram.

ITT indicates intention-to-treat.

Colchicine and MACE According to Atherosclerosis

In the ITT analysis, there were numerically fewer primary outcome events in the atherosclerosis group assigned to colchicine ($n=105$, 11.1%) compared with usual care ($n=132$, 14.2%). The corresponding rate of MACE in the colchicine arm was 3.88 (95% CI, 3.21–4.70) events/100 person-years versus 4.83 (95% CI,

4.07–5.73) events/100 person-years in the usual care arm in participants with atherosclerosis (Figure 2). After adjustment for mandatory randomization minimization variables and prior CAD, colchicine did not reduce the risk of the primary outcome in this group (adjusted HR [aHR], 0.83 [95% CI, 0.64–1.07]; Table 2). In patients without atherosclerosis, the number of primary outcome events was similar in the colchicine (48 [7.7%], 2.53 [95% CI, 1.91–3.36] events/100 person-years) and

Table 1. Baseline Characteristics According to Atherosclerosis Status

	Atherosclerosis (n=1875)	No atherosclerosis (n=1269)
Age, y	67.6 (9.5)	64.3 (10.3)
Female sex	546 (29.1)	407 (32.1)
Ethnicity		
White	1807 (96.4)	1194 (94.1)
Black	33 (1.8)	42 (3.3)
Asian	25 (1.3)	21 (1.7)
Other	10 (0.5)	12 (1.0)
Index event TIA	272 (14.5)	108 (8.5)
Hypertension	1310 (69.9)	747 (58.9)
Current smoker	462 (24.6)	232 (18.3)
Diabetes	474 (25.3)	227 (17.9)
Prior stroke	226 (12.1)	105 (8.3)
Prior CAD	366 (19.5)	0 (0)
Prior peripheral arterial disease	128 (6.8)	0 (0)
Prior carotid revascularization	165 (8.8)	0 (0)
Symptomatic luminal stenosis	615 (32.8)	0 (0)
Asymptomatic luminal stenosis	79 (4.2)	0 (0)
Nonstenosing plaque proximal to infarct	1050 (56.8)	0 (0)
LDL	2.68 (1.9–3.56)	2.90 (2.14–3.70)
CRP	3.1 (1.1–6.0)	2.9 (1.0–5.0)
modified Rankin Scale score	1 (0–2)	1 (0–2)
Statin therapy	1764 (98.1)	1168 (98.5)
Antiplatelet therapy	1840 (98.1)	1226 (96.6)

Data presented as mean (SD), median (interquartile intervals), or number (%) as appropriate. CAD indicates coronary artery disease; CRP, C-reactive protein; LDL, low-density lipoprotein; and TIA, transient ischemic attack.

usual care arms (53 [8.2%], 2.66 [95% CI, 2.03–3.49] events/100 person-years). In the nonatherosclerosis group, there was no statistically significant difference in the risk of MACE in the colchicine arm compared with usual care (aHR, 0.94 [95% CI, 0.64–1.39]; Figure 2; Table 2). In the ITT analysis, there was no evidence of interaction between colchicine and atherosclerosis for the primary outcome ($P_{\text{interaction}}=0.60$).

On prespecified on-treatment analysis, colchicine was associated with reduced MACE in patients with atherosclerosis (aHR, 0.76 [95% CI, 0.58–0.99]). By comparison, no association was observed between colchicine treatment and the primary outcome in participants without atherosclerosis (aHR, 0.95 [95% CI, 0.63–1.42]; Figure 3). However, there was no statistical evidence of treatment interaction ($P_{\text{interaction}}=0.37$).

Colchicine and Secondary Outcomes According to Atherosclerosis

There was no statistically significant difference between treatment arms in rates of the key secondary end point

or its individual components when stratified according to atherosclerosis subgroup (Table 2). However, the number of events was numerically fewer in patients with atherosclerosis assigned to colchicine versus usual care. There was a nonsignificant reduced risk of recurrent ischemic stroke in the colchicine arm (aHR, 0.76 [95% CI, 0.56–1.05]) that was observed in the atherosclerosis group only (aHR, 0.92 [95% CI, 0.60–1.42] in the non-atherosclerosis group; $P_{\text{interaction}}=0.48$; Figure S4).

Subgroup and Sensitivity Analysis

We explored the influence of sex on the effect of colchicine for MACE in patients with and without atherosclerosis. The results were consistent with the main analysis (Table S4). We also repeated our analysis excluding patients without vascular imaging who had been assigned to the nonatherosclerosis subgroup (n=212), and findings were similar for the primary outcome on both ITT and on-treatment analyses (Table S5).

DISCUSSION

In this secondary analysis of the CONVINCE trial, rates of major vascular events were substantially higher in participants with atherosclerosis compared with those without. Patients with atherosclerosis experienced an MACE event rate of $\approx 4.5\%/y$, representing a 66% relative risk increase and a 3-fold increased risk of coronary events when compared with patients without atherosclerosis. This association was observed irrespective of whether atherosclerosis was probably causal to the index stroke by TOAST criteria. We did not observe any evidence of treatment interaction for the effect of colchicine on vascular risk reduction according to atherosclerosis status in either the ITT or on-treatment analysis. However, patients with atherosclerosis who were assigned to colchicine had numerically fewer events than the usual care arm. After accounting for nonadherence in the on-treatment analysis, there was a statistically significant reduction in MACE for patients in the atherosclerosis group taking colchicine but not in patients without atherosclerosis.

Colchicine has several anti-inflammatory properties, including inhibition of the NLRP3 (NLR family pyrin domain-containing 3) inflammasome, microtubule assembly, neutrophil chemotaxis, and IL (interleukin)-1 β release.²¹ Several RCTs,^{22–24} but not all,^{25,26} have demonstrated efficacy for colchicine in the reduction of MACE and stroke in patients with CAD. Colchicine may mediate its protective effects on cardiovascular events through atherosclerotic plaque stabilization.²⁷ Inflammatory mechanisms play a key role in stroke pathogenesis and most of its subtypes.²⁸ For example, plaque inflammation quantified by carotid ¹⁸F-fluorodeoxyglucose positron emission tomography is strongly associated

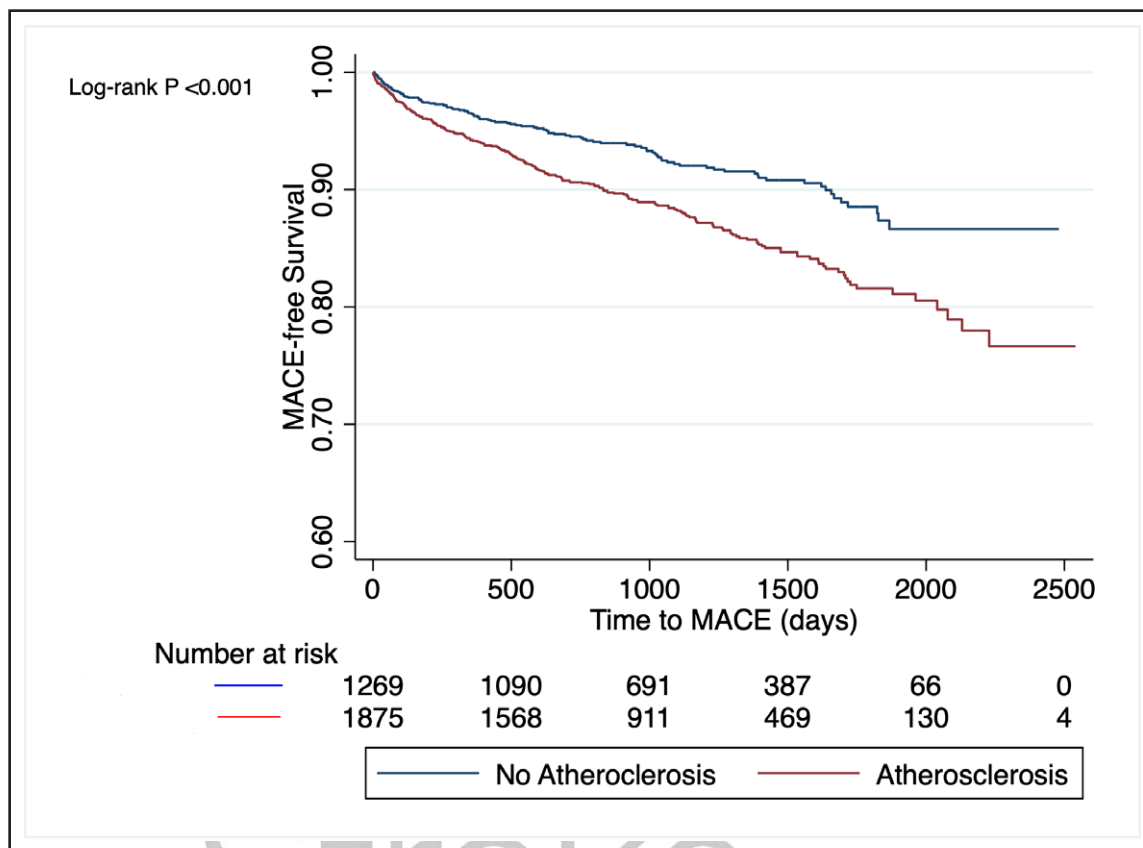


Figure 2. Risk of recurrent major adverse cardiovascular events (MACEs) stratified by atherosclerosis status.

with recurrent stroke in symptomatic carotid disease.²⁹ Mendelian randomization studies have shown that IL-6 and monocyte chemoattractant protein-1 signaling is causally associated with CAD and large artery stroke, with inconsistent or weaker evidence for lacunar or cardioembolic stroke.^{10,11} Histopathologic studies provide unequivocal evidence that inflammation drives plaque progression,^{4,5} and animal models implicate inflammasome activation in atrial remodeling and AF.³⁰ Genetic epidemiological data have also recently suggested a role for immune signaling in cardiac fibrosis and cardioembolism.¹² We have also recently shown that enhanced IL-6-CRP (C-reactive protein) signaling is implicated in vascular recurrence across a range of stroke subtypes, including cerebral small vessel disease and AF.^{31,32} Notwithstanding the complex role of immune pathways across stroke subtypes, individual patients frequently have multiple stroke-related disease traits that can each independently contribute to a residual vascular risk.¹³ For instance, atherosclerosis is frequently present in patients with AF¹⁴ and cerebral small vessel disease,³³ and nonstenosing plaque contributes significantly to cryptogenic stroke.³⁴ This complexity has led to considerable uncertainty and debate surrounding optimal patient selection for future RCTs of anti-inflammatory therapies for stroke prevention.³⁵ We have attempted to provide further clarification on

this key question by analyzing the efficacy of colchicine for vascular prevention in patients with and without atherosclerosis.

Our results provide important information for the design of future RCTs of anti-inflammatory therapies. First, rates of vascular recurrence were almost 70% higher in patients with atherosclerosis compared with those without, irrespective of whether a causal association with the qualifying stroke was likely. Second, rates of recurrent events were numerically fewer in patients assigned to colchicine versus usual care if they had atherosclerosis, but no efficacy signal was observed in those without atherosclerosis. Third, in the on-treatment analysis, patients who were adherent to colchicine had a statistically significantly lower risk of MACE in patients with atherosclerosis only, thereby suggesting a biological signal of efficacy for colchicine in these patients. Therefore, these results provide an important mechanistic link between immune mechanisms and poststroke recurrence in atherosclerosis. Taken together, these data provide justification for selecting patients with objective evidence of atherosclerosis in the next generation of anti-inflammatory prevention RCTs irrespective of whether atherosclerosis is thought to be causally relevant to the qualifying stroke. This approach would enrich the trial population by selecting patients with a considerably higher residual

Table 2. Primary and Secondary End Points in the Intention-to-Treat Population Stratified by Atherosclerosis Status

End point	Subgroup	Colchicine and usual care		Usual care		HR (95% CI)
		n (%)	Events per 100 person-years	n (%)	Events per 100 person-years	
Primary end point	Atherosclerosis	105 (11.1)	3.88 (3.21–4.70)	132 (14.2)	4.83 (4.07–5.73)	0.83 (0.64–1.07)
Primary end point	No atherosclerosis	48 (7.7)	2.53 (1.91–3.36)	53 (8.2)	2.66 (2.03–3.49)	0.94 (0.64–1.39)
Secondary end points						
Key secondary end point*	Atherosclerosis	99 (10.5)	3.66 (3.01–4.46)	124 (13.3)	4.54 (3.80–5.41)	0.82 (0.63–1.07)
Key secondary end point*	No atherosclerosis	48 (7.7)	2.53 (1.91–3.36)	53 (8.2)	2.66 (2.03–3.49)	0.94 (0.64–1.39)
Any ischemic stroke (fatal or nonfatal)	Atherosclerosis	68 (7.2)	2.51 (1.98–3.19)	91 (9.8)	3.33 (2.71–4.09)	0.76 (0.56–1.05)
Any ischemic stroke (fatal or nonfatal)	No atherosclerosis	40 (6.4)	2.11 (1.55–2.87)	45 (7.0)	2.26 (1.69–3.03)	0.92 (0.60–1.42)
Any major cardiac event (fatal or nonfatal)	Atherosclerosis	37 (3.9)	1.37 (0.99–1.89)	41 (4.4)	1.50 (1.10–2.04)	0.96 (0.61–1.49)
Any major cardiac event (fatal or nonfatal)	No atherosclerosis	8 (1.3)	0.42 (0.21–0.84)	8 (1.2)	0.40 (0.20–0.80)	1.05 (0.39–2.79)
Nonfatal ischemic stroke	Atherosclerosis	65 (6.9)	2.40 (1.88–3.06)	87 (9.3)	3.18 (2.58–3.93)	0.76 (0.55–1.05)
Nonfatal ischemic stroke	No atherosclerosis	38 (6.1)	2.00 (1.46–2.75)	44 (6.8)	2.21 (1.65–2.97)	0.90 (0.58–1.38)
Nonfatal MI	Atherosclerosis	22 (2.3)	0.81 (0.54–1.24)	25 (2.7)	0.91 (0.62–1.35)	0.93 (0.52–1.64)
Nonfatal MI	No atherosclerosis	4 (0.6)	0.21 (0.08–0.56)	4 (0.6)	0.20 (0.08–0.54)	1.08 (0.27–4.35)
Vascular death	Atherosclerosis	10 (1.1)	0.37 (0.20–0.69)	12 (1.3)	0.44 (0.25–0.77)	0.85 (0.37–1.97)
Vascular death	No atherosclerosis	6 (1.0)	0.32 (0.14–0.70)	5 (0.8)	0.25 (0.10–0.60)	1.24 (0.38–4.07)

HRs are adjusted for baseline minimization variables (age over 70 y, qualifying event, and time from event to randomization) and baseline CAD. In the atherosclerosis group, there were 943 patients in the intervention arm and 932 patients in the usual care arm. In the nonatherosclerosis group, there were 626 patients in the intervention arm and 643 in the usual care arm. CAD indicates coronary artery disease; HR, hazard ratio; and MI, myocardial infarction.

*Key secondary end point is defined as first nonfatal ischemic stroke, nonfatal myocardial infarction, nonfatal cardiac arrest, or vascular death.

vascular risk, thereby improving the statistical likelihood of observing a true treatment effect, while also selecting patients with a stronger biological plausibility of treatment response. This selection strategy might be complemented by selecting patients with elevated high-sensitivity CRP, which is consistently associated with an increased recurrence risk after stroke and has demonstrated reliability as a surrogate marker of treatment response.^{9,36} Finally, our results also demonstrate that the treatment benefit of colchicine appears to be sustained over years and underlines the potential need for long-term anti-inflammatory therapy to accrue benefit. This observation may also explain the absence of a signal for benefit in CHANCE-3, which focused on prevention of 90-day MACE, and emphasizes the need for long-term patient follow-up in the design of future trials.

Our study has limitations. Results did not show evidence of treatment interaction by atherosclerosis status on formal statistical testing, and colchicine may still have a similar effect size in patients without atherosclerosis. However, the sample size was not sufficiently powered to detect between-subgroup differences, and we intend that the present data should only be considered as hypothesis-generating. Assignment of atherosclerosis status was not centrally adjudicated, and we cannot entirely exclude the possibility of incorrect assignment. Trial participants were mostly White, so these findings may not be generalizable to other ethnic groups. Although this was a post hoc analysis, there is strong biological plausibility that treatment effects may differ in patients with or without atherosclerosis.

This work has strengths. First, we leveraged data from the only trial of anti-inflammatory therapy for the long-term prevention of MACE after stroke. Therefore, these data are important for informing the careful design of future RCTs. Second, rather than assigning participants into etiological subgroups according to the qualifying stroke mechanism alone, which underestimates the true burden of atherosclerosis, we defined atherosclerosis subgroups according to the presence of any evidence of clinically relevant plaque. Third, our data provide up-to-date evidence that the residual risk of major vascular recurrence after stroke in patients with atherosclerosis is considerable, even despite intensive and guideline-based best medical therapy. Our data suggest that the 5-year recurrence rates in this group are high and highlight an urgent need for more RCTs of novel strategies to reduce this residual risk.

CONCLUSIONS

In a post hoc analysis of the CONVINCe trial, there was no evidence of interaction between atherosclerosis and colchicine assignment for the primary outcome of MACE during follow-up in either the ITT or on-treatment population. However, event rates were numerically lower in the colchicine arm in patients with atherosclerosis, and a statistically significant benefit was observed in patients adherent to colchicine in this group. By contrast, no signal was seen in patients without atherosclerosis. Future RCTs of anti-inflammatory therapy for secondary stroke

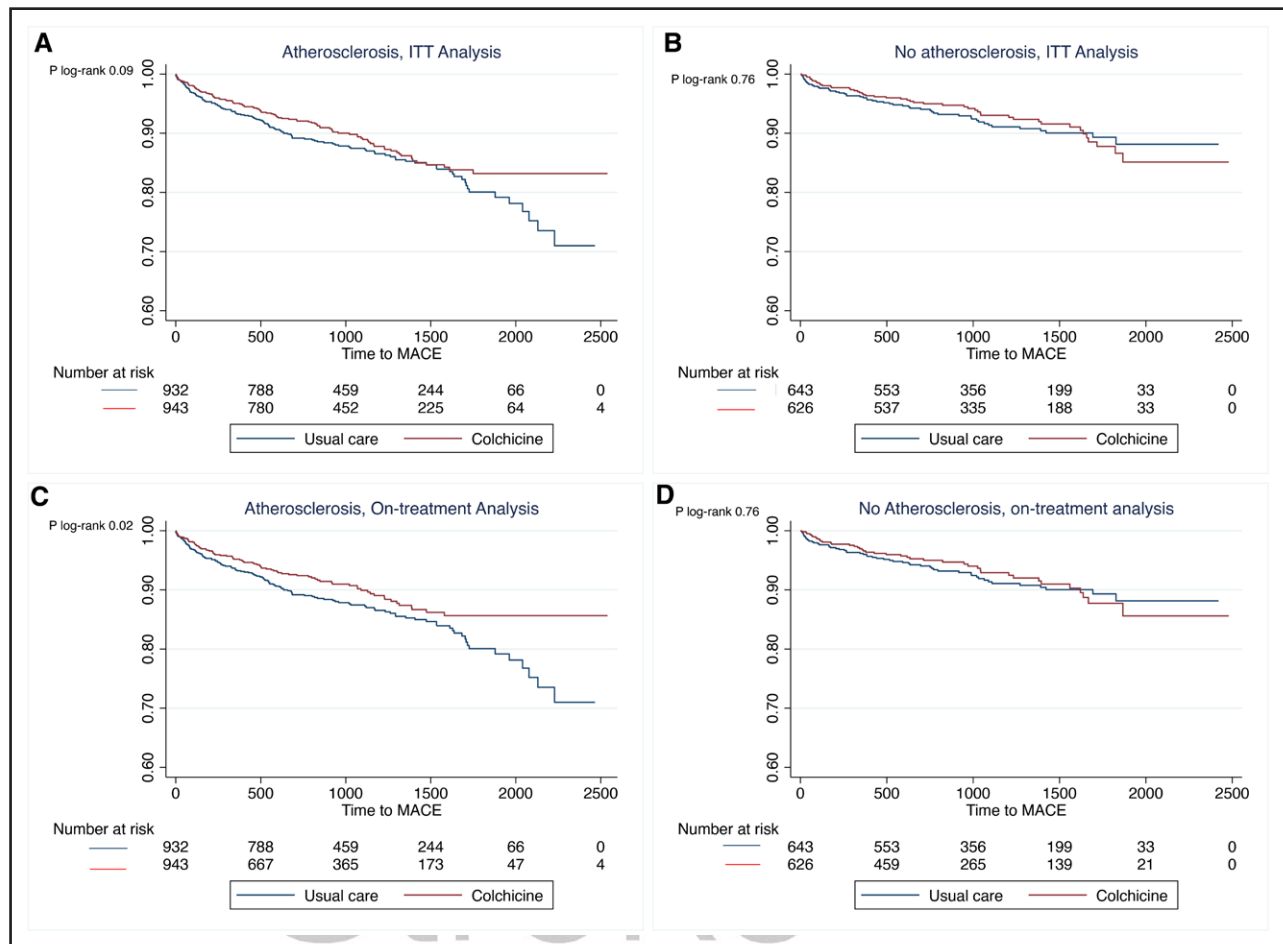


Figure 3. Cumulative incidence of the primary outcome stratified by atherosclerosis and randomized treatment assignment. **A**, Intention-to-treat (ITT) analysis in atherosclerosis subgroup. **B**, ITT analysis in patients without atherosclerosis. On-treatment analysis in **(C)** atherosclerosis and **(D)** nonatherosclerosis subgroups. MACE indicates major adverse cardiovascular event.

prevention should include patients with objective evidence of atherosclerosis, regardless of whether atherosclerosis is responsible for the recent stroke.

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Supplemental Material

Tables S1–S5
Figures S1–S4
STROBE Checklist
Trial Protocol

REFERENCES

- Feigin VL, Brainin M, Norrving B, Martins SO, Pandian J, Lindsay P, M FG, Rautalin I. World Stroke Organization: global stroke fact sheet 2025. *Int J Stroke*. 2025;20:132–144. doi: 10.1177/17474930241308142
- Boullanger M, Bejot Y, Rothwell PM, Touze E. Long-term risk of myocardial infarction compared to recurrent stroke after transient ischemic attack and ischemic stroke: systematic review and meta-analysis. *J Am Heart Assoc*. 2018;7:e007267. doi: 10.1161/JAHA.117.007267
- Libby P. Inflammation in atherosclerosis. *Nature*. 2002;420:868–874. doi: 10.1038/nature01323
- Marnane M, Prendeville S, McDonnell C, Noone I, Barry M, Crowe M, Mulligan N, Kelly PJ. Plaque inflammation and unstable morphology are associated with early stroke recurrence in symptomatic carotid stenosis. *Stroke*. 2014;45:801–806. doi: 10.1161/STROKEAHA.113.003657
- Redgrave JNE, Lovett JK, Gallagher RJ, Rothwell PM. Histological assessment of 526 symptomatic carotid plaques in relation to the nature and timing of ischemic symptoms: the Oxford plaque study. *Circulation*. 2006;113:2320–2328. doi: 10.1161/CIRCULATIONAHA.105.589044
- Nordestgaard AT, Moorthy MV, Cook NR, Rifai N, Lee IM, Buring JE, Ridker PM. High-sensitivity C-reactive protein, LDL cholesterol, lipoprotein(a) and 30-year risk of stroke in healthy women: a prospective, longitudinal cohort study. *Lancet Neurol*. 2025;24:920–930. doi: 10.1016/S1474-4422(25)00306-0
- Emerging Risk Factors C, Kaptoge S, Di Angelantonio E, Lowe G, Pepys MB, Thompson SG, Collins R, Danesh J. C-reactive protein concentration and risk of coronary heart disease, stroke, and mortality: an individual participant meta-analysis. *Lancet*. 2010;375:132–140. doi: 10.1016/S0140-6736(09)61717-7
- Papadopoulos A, Palaiopoulos K, Bjorkbacka H, Peters A, de Lemos JA, Seshadri S, Dichgans M, Georgakis MK. Circulating interleukin-6 levels and incident ischemic stroke: a systematic review and meta-analysis of prospective studies. *Neurology*. 2022;98:e1002–e1012. doi: 10.1212/WNL.00000000000013274
- McCabe JJ, Walsh C, Gorey S, Harris K, Hervella P, Iglesias-Rey R, Jern C, Li L, Miyamoto N, Montaner J, et al. C-reactive protein, interleukin-6, and vascular recurrence after stroke: an individual participant data meta-analysis. *Stroke*. 2023;54:1289–1299. doi: 10.1161/STROKEAHA.122.040529
- Zhang L, Omarov M, Xu L, deGoma E, Natarajan P, Georgakis MK. IL6 genetic perturbation mimicking IL-6 inhibition is associated with lower cardiometabolic risk. *Nat Cardiovasc Res*. 2025;4:1172–1186. doi: 10.1038/s44161-025-00700-7
- Georgakis MK, Gill D, Rannikmae K, Traylor M, Anderson CD, Lee JM, Kamatani Y, Hopewell JC, Worrall BB, Bernhagen J, et al. Genetically determined levels of circulating cytokines and risk of stroke. *Circulation*. 2019;139:256–268. doi: 10.1161/CIRCULATIONAHA.118.035905
- Brennan SO, Kelly PJ, Gorey S, Synnott P, Gill D, Dichgans M, Georgakis MK, Dib M-J, Gagnon E, Mahon N, et al. Genetically predicted IL-18 inhibition and risk of cardiovascular events: a Mendelian randomization study. *Circulation*. 2025;151:334–336. doi: 10.1161/CIRCULATIONAHA.124.072238
- Lavallée PC, Charles H, Albers GW, Caplan LR, Donnan GA, Ferro JM, Hennerici MG, Labreuche J, Molina C, Rothwell PM, et al; TIARegistry.org Investigators. Underlying causes of TIA and minor ischemic stroke and risk of major vascular events. *JAMA Neurol*. 2023;80:1199–1208. doi: 10.1001/jamaneurol.2023.3344
- Herlekar R, Sur RA, Hajiev S, Induruwa I, Agarwal S, Evans NR, Khadjoor K, Markus H, O'Brien E, Warburton E, et al. The contribution of competing mechanisms in stroke despite anticoagulation in patients with atrial fibrillation. *Eur Stroke J*. 2023;8:541–548. doi: 10.1177/23969873231168367
- Lavallée PC, Charles H, Albers GW, Caplan LR, Donnan GA, Ferro JM, Hennerici MG, Labreuche J, Molina C, Rothwell PM, et al; TIARegistry.org Investigators. Effect of atherosclerosis on 5-year risk of major vascular events in patients with transient ischaemic attack or minor ischaemic stroke: an international prospective cohort study. *Lancet Neurol*. 2023;22:320–329. doi: 10.1016/S1474-4422(23)00067-4
- Fiolet ATL, Poorthuis MHF, Opstal TSJ, Amarenco P, Boczar KE, Buysschaert I, Budgeon C, Chan NC, Cornel JH, Jolly SS, et al; Colchicine Cardiovascular Trialists Collaboration. Colchicine for secondary prevention of ischaemic stroke and atherosclerotic events: a meta-analysis of randomised trials. *eClinicalMedicine*. 2024;76:102835. doi: 10.1016/j.eclinm.2024.102835
- Kelly P, Lemmens R, Weimar C, Walsh C, Purroy F, Barber M, Collins R, Cronin S, Czlonkowska A, Desfontaines P, et al. Long-term colchicine for the prevention of vascular recurrent events in non-cardioembolic stroke (CONVINCE): a randomised controlled trial. *Lancet*. 2024;404:125–133. doi: 10.1016/S0140-6736(24)00968-1
- Li J, Meng X, Shi FD, Jing J, Gu HQ, Jin A, Jiang Y, Li H, Johnston SC, Hankey GJ, et al; CHANCE-3 Investigators. Colchicine in patients with acute ischaemic stroke or transient ischaemic attack (CHANCE-3): multicentre, double blind, randomised, placebo controlled trial. *BMJ*. 2024;385:e079061. doi: 10.1136/bmj-2023-079061
- Kelly P, Weimar C, Lemmens R, Murphy S, Purroy F, Arsovska A, Bornstein NM, Czlonkowska A, Fischer U, Fonseca AC, et al. Colchicine for prevention of vascular inflammation in Non-CardioEmbolic stroke (CONVINCE) - study protocol for a randomised controlled trial. *Eur Stroke J*. 2021;6:222–228. doi: 10.1177/2396987320972566
- von EE, Altman DG, Egger M, Pocock SJ, Gøtzsche PC, Vandenbroucke JP. Strengthening of Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. *BMJ*. 2007;335:806–808. doi: 10.1136/bmj.39335.541782.AD
- Deftereos SG, Beerkens FJ, Shah B, Giannopoulos G, Vrachatis DA, Giotaki SG, Siasos G, Nicolas J, Arnett C, Patel S, et al. Colchicine in cardiovascular disease: in-depth review. *Circulation*. 2022;145:61–78. doi: 10.1161/CIRCULATIONAHA.121.056171
- Nidorf SM, Eikelboom JW, Budgeon CA, Thompson PL. Low-dose colchicine for secondary prevention of cardiovascular disease. *J Am Coll Cardiol*. 2013;61:404–410. doi: 10.1016/j.jacc.2012.10.027
- Nidorf SM, Fiolet ATL, Mosterd A, Eikelboom JW, Schut A, Opstal TSJ, The SHK, Xu XF, Ireland MA, Lenderink T, et al; LoDoCo2 Trial Investigators. Colchicine in patients with chronic coronary disease. *N Engl J Med*. 2020;383:1838–1847. doi: 10.1056/NEJMoa2021372
- Tardif J-C, Kouz S, Waters DD, Bertrand OF, Diaz R, Maggioni AP, Pinto FJ, Ibrahim R, Gamra H, Kiwan GS, et al. Efficacy and safety of low-dose colchicine after myocardial infarction. *N Engl J Med*. 2019;381:2497–2505. doi: 10.1056/NEJMoa1912388
- Jolly SS, d'Entremont M-A, Lee SF, Mian R, Tyrwhitt J, Kedev S, Montalescot G, Cornel JH, Stanković G, Moreno R, et al; CLEAR Investigators. Colchicine in acute myocardial infarction. *N Engl J Med*. 2025;392:633–642. doi: 10.1056/NEJMoa2405922
- Tong DC, Quinn S, Nasis A, Hiew C, Roberts-Thomson P, Adams H, Sriamameswaran R, Htun NM, Wilson W, Stub D, et al. Colchicine in patients with acute coronary syndrome: the Australian COPS randomized clinical trial. *Circulation*. 2020;142:1890–1900. doi: 10.1161/CIRCULATIONAHA.120.050771
- Yu M, Yang Y, Dong SL, Zhao C, Yang F, Yuan YF, Liao YH, He SL, Liu K, Wei F, et al. Effect of colchicine on coronary plaque stability in acute coronary syndrome as assessed by optical coherence tomography: the COLOCT randomized clinical trial. *Circulation*. 2024;150:981–993. doi: 10.1161/CIRCULATIONAHA.124.069808
- Zietz A, Gorey S, Kelly PJ, Katan M, McCabe JJ. Targeting inflammation to reduce recurrent stroke. *Int J Stroke*. 2024;19:379–387. doi: 10.1177/17474930231207777
- McCabe JJ, Camps-Renom P, Giannotti N, McNulty JP, Coveney S, Murphy S, Barry M, Harbison J, Cronin S, Williams D, et al. Carotid plaque inflammation imaged by PET and prediction of recurrent stroke at 5 years. *Neurology*. 2021;97:e2282–e2291. doi: 10.1212/WNL.00000000000012909

30. Yao C, Veleza T, Scott L Jr, Cao S, Li L, Chen G, Jeyabal P, Pan X, Alsina KM, Abu-Taha ID, et al. Enhanced cardiomyocyte NLRP3 inflammasome signaling promotes atrial fibrillation. *Circulation*. 2018;138:2227–2242. doi: 10.1161/CIRCULATIONAHA.118.035202
31. McCabe JJ, Walsh C, Gorey S, Harris K, Hervella P, Iglesias-Rey R, Jern C, Li L, Miyamoto N, Montaner J, et al. C-reactive protein, interleukin-6, and vascular recurrence according to stroke subtype: an individual participant data meta-analysis. *Neurology*. 2024;102:e208016. doi: 10.1212/WNL.000000000000208016
32. McCabe JJ, Harris K, Gorey S, Walsh C, Arnold M, De Marchis GM, Hervella P, Iglesias-Rey R, Jern C, Katan M, et al. Systemic inflammation and recurrence after atrial fibrillation-related stroke: an individual participant data meta-analysis. *Neurology*. 2026;106:e214701. doi: 10.1212/WNL.000000000000214701
33. Gongora-Rivera F, Labreuche J, Jaramillo A, Steg PG, Hauw J-J, Amarenco P. Autopsy prevalence of coronary atherosclerosis in patients with fatal stroke. *Stroke*. 2007;38:1203–1210. doi: 10.1161/01.STR.0000260091.13729.96
34. Kamtchum-Tatuene J, Wilman A, Saqqur M, Shuaib A, Jickling GC. Carotid plaque with high-risk features in embolic stroke of undetermined source: systematic review and meta-analysis. *Stroke*. 2020;51:311–314. doi: 10.1161/STROKEAHA.119.027272
35. Georgakis MK, Melton P, Zivkovic L, Kopczak A, Katsanos AH. Anti-inflammatory therapies for atherosclerotic stroke prevention. *Neurology*. 2025;105:e214214. doi: 10.1212/WNL.000000000000214214
36. 11th European Stroke Organisation Conference Abstracts – 21-23 May 2025, Helsinki, Finland. *Eur Stroke J*. 2025;10:3–789. doi: 10.1177/23969873251329993



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